

The Cosmological Lithium Problem and the CPTG Lithium Solution

Carter L Glass Jr

Abstract

The cosmological lithium problem is the selective failure of standard big-bang nucleosynthesis to reproduce the low primordial mass-seven abundance inferred from old metal-poor stellar systems while retaining successful deuterium and helium agreement. Curvature Polarization Transport Gravity, or CPTG, treats this as a surviving mass-seven transport problem inside the primordial reaction network. The proposed CPTG correction is a locked geometric transport gate with optical depth fixed to $\ln \pi$, acting directly on the live ${}^7\text{Li}$ and ${}^7\text{Be}$ channels. The resulting survival factor is $1/\pi$, with no fitted depletion coefficient. The abundance coordinate is the CPTG transported BBN coordinate, not the acoustic/CMB baryon coordinate. Under this rule, the reaction-network controls give $10^6\text{D}/\text{H} = 25.074383114428$ and $Y_p = 0.245694338195$, both within the observational control bands, while the locked mass-seven gate moves raw high lithium into the observed lithium band. The source-network validation gives 205/205 passing PRy-Mordial background-admissible cases, 24/24 passing AlterBBN thermal-window and shape cases, 33/33 passing AlterBBN eta/rate admitted gated rows, and 484/484 passing background-admissible gated rows in the AlterBBN $N = 200$ combined Monte Carlo audit. The result is a two-code source-network validation of the locked CPTG $A = 7$ transport gate within the stated BBN-background admissibility standard.

1 The Cosmological Lithium Problem

Big-bang nucleosynthesis is a high-precision early-universe test because light-element yields respond directly to the baryon-to-photon ratio, the relativistic radiation content, the neutron-proton conversion history, and the thermonuclear reaction network. Standard calculations can reproduce deuterium and helium at high precision, but they overpredict the primordial abundance of mass seven relative to the lithium plateau inferred from old metal-poor stars [1, 2, 3].

The observational issue is not a uniform failure of light-element nucleosynthesis. It is a selective mass-seven excess. Most of the final primordial ${}^7\text{Li}$ is carried through ${}^7\text{Be}$ during the nuclear epoch and later appears as lithium after electron capture. The relevant network quantity is therefore the surviving mass-seven abundance:

$$Y_7 = Y_{7\text{Li}} + Y_{7\text{Be}}. \quad (1)$$

The lithium comparison anchor used in this calculation is

$${}^7\text{Li}/\text{H}_{\text{obs}} = (1.45 \pm 0.25) \times 10^{-10}. \quad (2)$$

The corresponding deuterium and helium controls are

$$\text{D}/\text{H}_{\text{obs}} = (25.08 \pm 0.29) \times 10^{-6}, \quad (3)$$

and

$$Y_{p,\text{obs}} = 0.245 \pm 0.003. \quad (4)$$

A viable CPTG lithium solution must therefore satisfy three simultaneous conditions: preserve the successful deuterium control, preserve the successful helium control, and reduce the surviving mass-seven abundance without introducing a free lithium depletion factor.

2 CPTG Native Ordering

CPTG is organized as a geometric theory in which baryonic matter sources nonlinear curvature polarization and curvature transport. Its scalar laws are reductions of a native nonlinear geometric branch, not independent fitting rules imposed separately in each observational sector [11, 12].

The schematic CPTG action is

$$S_{\text{CPTG}} = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R + \mathcal{L}_{\text{pol}} + \mathcal{L}_{\text{trans}} + \mathcal{L}_m \right]. \quad (5)$$

Variation with respect to the metric gives the schematic field equation

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (6)$$

The comparison hierarchy is

$$\begin{array}{c} \text{CPTG-native geometric quantity} \\ \downarrow \\ \text{comparison-coordinate quantity} \\ \downarrow \\ \text{reported observational quantity.} \end{array} \quad (7)$$

For primordial abundances, this becomes

$$\begin{array}{c}
\text{native curvature transport} \\
\downarrow \\
\text{transported BBN abundance coordinate} \\
\downarrow \\
\text{live lithium source-network gate} \\
\downarrow \\
\text{observed light-element abundances.}
\end{array} \tag{8}$$

This ordering is essential. The lithium gate is not a post-processing abundance rescaling. It is a source-network realization of the curvature-transport sector acting on the live abundance variables before the final mass-seven comparison is made.

3 BBN Coordinate and Light-Element Conversion

The CPTG pi branch separates the acoustic/CMB baryon coordinate from the nuclear reaction-network coordinate. The native curvature exponent is

$$p_C = \pi. \tag{9}$$

The acoustic exponent is

$$p_{ac} = 3 - \frac{\pi}{100} = 2.968584073464. \tag{10}$$

The acoustic transport factor is

$$G_T = \frac{p_{ac}}{p_C} = 0.9449296585513721. \tag{11}$$

The acoustic baryon-to-photon coordinate is

$$\eta_{10}^{ac} = 6.170380167710. \tag{12}$$

The BBN abundance coordinate is obtained by the transport-scaled conversion

$$\eta_{10}^{BBN} = \eta_{10}^{ac} \sqrt{G_T}. \tag{13}$$

Numerically,

$$\eta_{10}^{BBN} = 5.998071834744. \tag{14}$$

The corresponding physical baryon-density coordinate is

$$\Omega_b h_{BBN}^2 = 0.021898765370. \tag{15}$$

The effective radiation comparison value is identified with the acoustic exponent:

$$N_{\text{eff}}^{\text{CPTG}} = p_{\text{ac}} = 2.968584073464. \quad (16)$$

Using the transported BBN coordinate, the light-element control outputs are

$$Y_{p,\text{network}} = 0.245694338195, \quad (17)$$

and

$$10^6(\text{D/H})_{\text{network}} = 25.074383114428. \quad (18)$$

The acoustic/CMB coordinate is therefore not inserted directly into the nuclear abundance comparison. The abundance test is performed in the transported BBN coordinate, with D/H and Y_p acting as the background controls.

4 Locked Mass-Seven Transport Gate

The CPTG lithium mechanism is a fixed optical-depth gate on the surviving mass-seven network channels. The final mass-seven abundance is

$$Y_{7,\text{final}} = Y_{7,\text{raw}} \exp \left[- \int \Gamma_{7,\text{trans}}(t) dt \right]. \quad (19)$$

The CPTG geometric condition is

$$\int \Gamma_{7,\text{trans}}(t) dt = \ln \pi. \quad (20)$$

Therefore,

$$Y_{7,\text{CPTG}} = Y_{7,\text{raw}} e^{-\ln \pi} = \frac{Y_{7,\text{raw}}}{\pi}. \quad (21)$$

The locked survival factor is

$$\frac{1}{\pi} = 0.3183098861837907. \quad (22)$$

At source-network level, the live lithium channel is evolved with

$$\frac{dY_{7\text{Li}}}{dt} \longrightarrow \left(\frac{dY_{7\text{Li}}}{dt} \right)_{\text{BBN}} - \Gamma_{7,\text{trans}} Y_{7\text{Li}}. \quad (23)$$

The live beryllium channel is evolved with

$$\frac{dY_{7\text{Be}}}{dt} \longrightarrow \left(\frac{dY_{7\text{Be}}}{dt} \right)_{\text{BBN}} - \Gamma_{7,\text{trans}} Y_{7\text{Be}}. \quad (24)$$

For a thermal gate, the transport rate is written as

$$\Gamma_{7,\text{trans}}(T) = \Gamma_0 W_7[T(t)]. \quad (25)$$

The normalization is always locked by

$$\int \Gamma_0 W_7[T(t)] dt = \ln \pi. \quad (26)$$

The gate shape and thermal support may be varied as robustness controls, but the integrated optical depth is not fitted.

5 Observed Controls and Admissibility Rule

For each observable X , the signed pull is

$$\Delta_X = \frac{X_{\text{CPTG}} - X_{\text{obs}}}{\sigma_X}. \quad (27)$$

The abundance-screening gate used here is

$$|\Delta_X| < 2. \quad (28)$$

A background is admitted only if the un-gated background already satisfies the deuterium and helium controls:

$$|\Delta_{\text{D/H}}| < 2, \quad |\Delta_{Y_p}| < 2. \quad (29)$$

Lithium is not used for background admission because the raw lithium excess is the target of the test. Once the background is admitted, the locked mass-seven gate is required to satisfy

$$|\Delta_{7\text{Li/H}}| < 2. \quad (30)$$

Observable	Observed value	CPTG or control value	Pull
D/H	$(25.08 \pm 0.29) \times 10^{-6}$	$25.074383114428 \times 10^{-6}$	-0.019σ
Y_p	0.245 ± 0.003	0.245694338195	$+0.231\sigma$
$^7\text{Li/H}$, PRyMordial raw	$(1.45 \pm 0.25) \times 10^{-10}$	5.4386×10^{-10}	$+15.95\sigma$
$^7\text{Li/H}$, AlterBBN raw	$(1.45 \pm 0.25) \times 10^{-10}$	5.376×10^{-10}	$+15.704\sigma$
$^7\text{Li/H}$, representative locked gate	$(1.45 \pm 0.25) \times 10^{-10}$	1.713×10^{-10}	$+1.052\sigma$

Table 1: Observed abundance anchors and representative CPTG source-network comparisons. The raw mass-seven rows show the lithium problem. The locked-gate row shows the source-network effect of the geometric $\ln \pi$ optical depth.

6 Source-Network Implementation

The calculation was tested in two independent BBN source-network environments: PRyMordial and AlterBBN. PRyMordial is a modern early-universe BBN package designed for light-element abundance and effective-radiation calculations [6]. AlterBBN is an independent public C code for BBN abundance calculations in standard and nonstandard cosmological backgrounds [7, 8, 9].

In PRyMordial, the final mass-seven abundance is evaluated from the live ${}^7\text{Li}$ and ${}^7\text{Be}$ channels. The CPTG gate is therefore inserted into both channels before the final abundance is read.

In AlterBBN, the live network indices used by the source patch are

$$Y_8 = Y_{7\text{Li}}, \quad Y_9 = Y_{7\text{Be}}. \quad (31)$$

The patched live-channel evolution is

$$\begin{aligned} \frac{dY_8}{dt} &\longrightarrow \frac{dY_8}{dt} - \Gamma_7(T)Y_8, \\ \frac{dY_9}{dt} &\longrightarrow \frac{dY_9}{dt} - \Gamma_7(T)Y_9. \end{aligned} \quad (32)$$

AlterBBN reports final mass-seven lithium through its own post-processing convention, in which the ${}^7\text{Be}$ contribution is already carried into the final ${}^7\text{Li}/\text{H}$ output. The reported ${}^7\text{Be}/\text{H}$ diagnostic column is therefore not added a second time.

7 BBN Table Controls

CAMB BBN-table interpolation was used only as a D/H and helium coordinate control, not as a live lithium source-network driver. CAMB documents BBN table interpolation using PArthENoPE and PRIMAT-family tables [10, 4, 5]. The CPTG BBN-coordinate row gives the PRIMAT-family D/H control

$$10^6(\text{D}/\text{H})_{\text{PRIMAT,BBN}} = 25.0746. \quad (33)$$

The corresponding pull is

$$\Delta_{\text{D}/\text{H}} = -0.019\sigma. \quad (34)$$

This confirms that the transported BBN coordinate is compatible with the deuterium control before the live mass-seven transport gate is judged.

8 Two-Code Source-Network Results

The source-network results are summarized in Table 2. The central feature is that the raw lithium abundance is high in both independent networks, while the same locked $\ln \pi$ optical depth moves admitted cases into the observed lithium band without spoiling deuterium or helium controls.

Code layer	Control or stress class	D/H range	Locked ${}^7\text{Li}/\text{H}$ range	Pass count
PRyMordial	Background-admissible source-network Monte Carlo	$(24.5342\text{--}25.6310) \times 10^{-6}$	$(1.5820\text{--}1.7883) \times 10^{-10}$	205/205
AlterBBN	Thermal-window and gate-shape scan	24.63×10^{-6}	$(1.712\text{--}1.719) \times 10^{-10}$	24/24
AlterBBN	Eta/rate admissible rows	$(24.63\text{--}25.55) \times 10^{-6}$	$(1.602\text{--}1.768) \times 10^{-10}$	33/33
AlterBBN	200-sample combined eta/window/shape Monte Carlo	$(24.51\text{--}25.65) \times 10^{-6}$	$(1.571\text{--}1.790) \times 10^{-10}$	484/484

Table 2: Two-code source-network validation of the locked CPTG mass-seven transport gate. D/H is shown in the same units as the observational anchor. Lithium ranges are final mass-seven lithium comparisons.

The best PRyMordial geometric survival-ratio match is

$$\frac{Y_{7,\text{gated}}}{Y_{7,0}} = 0.3183117923. \quad (35)$$

Its absolute difference from the locked value is

$$\left| \frac{Y_{7,\text{gated}}}{Y_{7,0}} - \frac{1}{\pi} \right| = 1.91 \times 10^{-6}. \quad (36)$$

The best AlterBBN combined Monte Carlo admitted row gives

$$\frac{Y_{7,\text{gated}}}{Y_{7,0}} = 0.3182629387. \quad (37)$$

Its signed difference from the locked value is

$$\frac{Y_{7,\text{gated}}}{Y_{7,0}} - \frac{1}{\pi} = -4.6947 \times 10^{-5}. \quad (38)$$

The combined AlterBBN admitted set has median absolute ratio residual

$$\text{median} \left| \frac{Y_{7,\text{gated}}}{Y_{7,0}} - \frac{1}{\pi} \right| = 4.7200 \times 10^{-4}. \quad (39)$$

The maximum absolute ratio residual in that admitted set is

$$\max \left| \frac{Y_{7,\text{gated}}}{Y_{7,0}} - \frac{1}{\pi} \right| = 1.9454 \times 10^{-3}. \quad (40)$$

The final AlterBBN combined Monte Carlo contains 206 total samples including anchors, 824 baseline rows, and 824 gated rows. Background admission gives

$$824 = 484_{\text{admitted}} + 340_{\text{rejected}}. \quad (41)$$

All admitted gated rows pass the declared D/H, Y_p , and lithium gates:

$$\frac{484}{484} = 1.000. \quad (42)$$

The rejected rows are rejected before the lithium gate is judged because the background D/H control already fails; their rejection is therefore a background-coordinate control result, not a lithium-gate failure.

The finite offset from the exact $1/\pi$ ratio in AlterBBN is consistent with output precision and grid-level source integration. The normalization itself remains fixed to $\ln \pi$ in every tested row.

9 Direct Observed-versus-CPTG Outcome

The final observed-versus-CPTG comparison is shown in Table 3. The table uses the declared observational anchors and reports the abundance-sector result without fitting the lithium survival factor.

Observable	Observed value	CPTG result	Residual	Result
D/H	$(25.08 \pm 0.29) \times 10^{-6}$	$25.074383114428 \times 10^{-6}$	-0.019σ	Pass
Y_p	0.245 ± 0.003	0.245694338195	$+0.231\sigma$	Pass
${}^7\text{Li}/\text{H}$, raw network	$(1.45 \pm 0.25) \times 10^{-10}$	$(5.376\text{--}5.4386) \times 10^{-10}$	$(+15.704\text{--}15.95)\sigma$ +	Fail
${}^7\text{Li}/\text{H}$, locked CPTG gate	$(1.45 \pm 0.25) \times 10^{-10}$	$(1.571\text{--}1.790) \times 10^{-10}$	within 2σ	Pass

Table 3: Direct observed-versus-CPTG abundance comparison. The raw network row displays the lithium problem. The locked CPTG gate row displays the final admitted source-network outcome.

The result can be written compactly as

$$\begin{aligned} &\text{raw high mass seven} \longrightarrow \text{locked } \tau_7 = \ln \pi \text{ source-network gate} \\ &\longrightarrow \text{observed lithium band with D/H and } Y_p \text{ preserved.} \end{aligned} \quad (43)$$

No baryon-coordinate retune is used in this step. No fitted lithium depletion factor is introduced. No post-network division is applied after the abundance calculation. The live mass-seven abundance is evolved through the locked gate inside the source network.

10 Interpretation

The CPTG interpretation is that primordial lithium exposes a late mass-seven survival channel rather than a failure of the entire light-element abundance system.

Deuterium and helium set the admitted BBN background. The locked CPTG transport gate then acts selectively on the live $A = 7$ channels during a late, cool network interval where D/H and Y_p are already frozen to their observed-control values.

This explains why the same abundance coordinate can pass D/H and helium while requiring an additional geometric transport effect in the mass-seven sector. The successful rows are not accepted because lithium was tuned into agreement. They are accepted because the un-gated background first passes D/H and helium, and the fixed $\ln \pi$ gate then carries the raw high mass-seven abundance into the observed lithium range.

The controlling distinction is therefore not between lithium success and deuterium failure. It is between BBN-background-admissible rows and rows whose D/H or Y_p controls already fail before the lithium gate is judged. Within the admitted set, the locked mass-seven survival factor is stable across gate shapes, thermal windows, eta/rate envelopes, and the 200-sample combined AlterBBN Monte Carlo sampling.

11 Validation Scope and Falsifiability

The result establishes a two-code source-network validation of the locked CPTG $A = 7$ gate under the stated admissibility rule. The scope is precise. It is a validation of the fixed $\ln \pi$ source-network transport mechanism in PRyMordial and AlterBBN, with D/H and helium used as independent background controls. The final AlterBBN combined Monte Carlo layer contains 484/484 passing admitted gated rows after D/H -background admission.

The mechanism is falsifiable. It fails if the lithium agreement requires replacing $\ln \pi$ by a fitted optical depth, if the live ${}^7\text{Li}$ and ${}^7\text{Be}$ source terms cannot reproduce the $1/\pi$ survival behavior in an admissible network background, if D/H or helium fails on the admitted abundance coordinate, or if the result depends on adding the AlterBBN diagnostic ${}^7\text{Be}/H$ column twice.

The successful test condition is

$$\begin{aligned} |\Delta_{D/H}| < 2, \quad |\Delta_{Y_p}| < 2, \\ |\Delta_{{}^7\text{Li}/H}| < 2, \quad \int \Gamma_{7,\text{trans}} dt = \ln \pi. \end{aligned} \tag{44}$$

The validation does not require a universal claim about every possible BBN implementation. It states that two independent source-network codes, tested under a fixed optical depth and a fixed BBN-coordinate rule, reproduce the CPTG mass-seven survival prediction while preserving the deuterium and helium controls.

12 Conclusion

The cosmological lithium problem is a selective mass-seven excess: standard source networks can preserve D/H and helium agreement while producing too much final lithium. CPTG resolves this within the tested source-network scope by applying a locked geometric transport gate to the live ${}^7\text{Li}$ and ${}^7\text{Be}$ channels.

The governing relation is

$$Y_{7,\text{CPTG}} = \frac{Y_{7,\text{raw}}}{\pi}, \quad \tau_{7,\text{trans}} = \ln \pi. \quad (45)$$

The transported BBN coordinate is

$$\eta_{10}^{\text{BBN}} = 5.998071834744, \quad \Omega_b h_{\text{BBN}}^2 = 0.021898765370. \quad (46)$$

The light-element controls are

$$10^6(\text{D/H})_{\text{network}} = 25.074383114428, \quad Y_{p,\text{network}} = 0.245694338195. \quad (47)$$

The source-network validation result is

$$205/205, \quad 24/24, \quad 33/33, \quad 484/484. \quad (48)$$

Thus, within the stated two-code source-network standard, the CPTG $A = 7$ lithium gate is validated: the fixed $\ln \pi$ transport optical depth converts raw high mass-seven output into the observed lithium band while preserving the D/H and helium controls.

References

- [1] M. Spite and F. Spite, “Lithium abundance at the formation of the Galaxy,” *Nature* **297**, 483–485 (1982).
- [2] R. H. Cyburt, B. D. Fields, K. A. Olive, and T.-H. Yeh, “Big bang nucleosynthesis: Present status,” *Reviews of Modern Physics* **88**, 015004 (2016).
- [3] B. D. Fields, P. Molaro, and S. Sarkar, “Big-Bang Nucleosynthesis,” Particle Data Group review, revised August 2025. <https://ccwww.kek.jp/pdg/2025/reviews/rpp2025-rev-bbang-nucleosynthesis.pdf>.
- [4] C. Pitrou, A. Coc, J.-P. Uzan, and E. Vangioni, “Precision big bang nucleosynthesis with improved Helium-4 predictions,” *Physics Reports* **754**, 1–66 (2018).

- [5] S. Gariazzo et al., “PArthENoPE revolutions,” *Computer Physics Communications* **271**, 108205 (2022).
- [6] A.-K. Burns, T. M. P. Tait, and M. Valli, “PRyMordial: The First Three Minutes, Within and Beyond the Standard Model,” *European Physical Journal C* **84**, 86 (2024).
- [7] A. Arbey, “AlterBBN: A program for calculating the BBN abundances of the elements in alternative cosmologies,” *Computer Physics Communications* **183**, 1822–1831 (2012), arXiv:1106.1363.
- [8] A. Arbey, J. Auffinger, K. P. Hickerson, and E. S. Jenssen, “AlterBBN v2: A public code for calculating Big-Bang nucleosynthesis constraints in alternative cosmologies,” *Computer Physics Communications* **248**, 106982 (2020), arXiv:1806.11095.
- [9] AlterBBN project homepage, “AlterBBN: A program which computes the abundances of the elements predicted by Big-Bang nucleosynthesis.” <https://alterbbn.hepforge.org/>.
- [10] CAMB documentation, “BBN models,” including PRIMAT and PArthENoPE table interpolation support. <https://camb.readthedocs.io/en/latest/bbn.html>.
- [11] Carter L Glass Jr, Curvature Polarization Transport Gravity: A Unified Geometric Framework for Cosmic Structure and Expansion, DOI: 10.13140/RG.2.2.26030.68164.
- [12] Carter L Glass Jr, CPTG Geometric Pi Branch: Comparison Coordinates, DOI: 10.13140/RG.2.2.15964.35201.
- [13] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at <https://github.com/CLG2025/CPTG>.